# HANGOO KANG

*AI Researcher — LLM Post-Training, Agentic AI, MLSys*
Stanford, CA • hangook@stanford.edu • (217) 904-9718 • Google Scholar

## EDUCATION

**Stanford University** *Aug 2025 — May 2027*
*M.S. in Computer Science (AI track) | GPA: 3.9 / 4.00* Stanford, CA
Advised by Profs. Azalia Mirhoseini and Yejin Choi.

**University of Illinois Urbana–Champaign** *Aug 2021 — May 2025*
*B.S. in Computer Science | GPA: 3.96 / 4.00* Urbana, IL
Dean's List: Spring 2022, Spring 2023, Fall 2023, Spring 2024.

## SELECTED PUBLICATIONS ** equal contribution*

**[1] ThunderSyncRL: Lossless Acceleration of Agentic Reinforcement Learning.**
Seil Kang*, **Hangoo Kang***, Tarun Suresh, Youngeun Kim, Shreyas Pimpalgaonkar, Seong Jae Hwang, Azalia Mirhoseini.
***arXiv preprint, 2026***. paper · code

**[2] Sharpening Tax in Post-Training.**
Changdae Oh, Qi Zeng, Qi Qi, Andrey Zhmoginov, Deren Lei, Yun He, Hoang Phan, **Hangoo Kang**, Azalia Mirhoseini, Sharon Li.
***arXiv preprint, 2026***. paper · code

**[3] Block Parallelism for Efficient Distributed Long-Context Diffusion Language Model Training.**
Tarun Suresh*, Pranshu Chaturvedi*, **Hangoo Kang***, Parth Shroff, Ishan S. Khare, Hermann Kumbong, Azalia Mirhoseini.
***arXiv preprint, 2026***. paper · code

**[4] Training Hybrid Block Diffusion Language Models with Partial Bidirectionality.**
Pranshu Chaturvedi, Parth Shroff, Tarun Suresh, **Hangoo Kang**, Kaiyue Wen.
***arXiv preprint, 2026***. paper

**[5] Decentralized Multi-Agent Systems with Shared Context.**
Yuzhen Mao, Jerry Gu, Aadi Chauhan, Qizheng Zhang, **Hangoo Kang**, Azalia Mirhoseini.
***arXiv preprint, 2026***. paper · project

**[6] TRACE: Capability-Targeted Agentic Training.**
**Hangoo Kang***, Tarun Suresh*, Jon Saad-Falcon, Azalia Mirhoseini.
***AIWILD @ ICML 2026 (Spotlight)***. paper · code

**[7] Forty Shades of Blue: Quality-Diversity Alignment via Mode-Conditioned Reinforcement Learning.**
Jiayi Yuan*, **Hangoo Kang***, James Jihao Liu*, Yejin Choi, Vikram Iyer, Liwei Jiang, Natasha Jaques.
***GenAICreativity @ ICML 2026***.

**[8] TRAP: Targeted Redirecting of Agentic Preferences.**
**Hangoo Kang***, Jehyeok Yeon*, Gagandeep Singh.
***NeurIPS 2025*** paper · code

**[9] Learning a Pessimistic Reward in RLHF: KL Regularization is Not Necessary.**
Yinglun Xu*, **Hangoo Kang***, Tarun Suresh, Yuxuan Wan, Gagandeep Singh.
***ARLET @ NeurIPS 2025*** paper

**[10] Stochastic Monkeys at Play: Random Augmentations Cheaply Break LLM Safety Alignment.**
Jason Vega, **Hangoo Kang***, Junsheng Huang*, Gaokai Zhang*, Minjia Zhang, Gagandeep Singh.
***Under review.*** paper · code

**[11] SynCode: LLM Generation with Grammar Augmentation.**
Shubham Ugare, Tarun Suresh, **Hangoo Kang**, Sasa Misailovic, Gagandeep Singh.
***TMLR 2025*** paper · code

## RESEARCH EXPERIENCE

**Stanford University — Scaling Intelligence Lab** *Aug 2025 — Present*

***Fully Funded Research Assistant*** · *Advised by Prof. Azalia Mirhoseini* Stanford, CA

- Lead author of *TRACE* (AIWILD @ ICML 2026): end-to-end system for capability-targeted agent self-improvement that contrasts successful and failed trajectories, synthesizes per-capability training environments, and trains LoRA adapters via GRPO, achieving +14.1 pts on τ²-Bench and +9.2 pts over GRPO at matched rollout budget.
- Co-author on PyTorch / Meta Superintelligence Lab production blog releases on *torchforge* (PyTorch-native scalable RL post-training): Introducing TorchForge and Scalable RL with TorchForge and Weaver; contributor to the open-source meta-pytorch/torchforge repository.
- Drove the Forge × Weaver verifier integration validated on a 512-GPU CoreWeave cluster, reaching 4× faster RL step times, >90% job completion, and matching o3-mini on MATH500 / GPQA / MMLU-Pro without human annotations.
- Co-leading the universal verifier project for agentic tasks to avoid domain specific post training with low compute requirement.

**Stanford University — xLab** *Aug 2025 — Present*

*Graduate Researcher · Advised by Prof. Yejin Choi (Stanford) & Prof. Natasha Jaques (UW)* Stanford, CA

- Co-first author on *MODA*, investigating model collapse and inter-model homogeneity (the "Artificial Hivemind" effect) in frontier LMs (LLaMA-3, Gemini-2.5, GPT-4o), and designing multi-agent RL training to recover diverse, human-like generation distributions.
- Building an automated synthetic data generation framework for LLM alignment in out-of-distribution and underspecified domains.

**UIUC — FOCAL Lab & ARC Group** *Spring 2023 — Spring 2025*

*Undergraduate Researcher · Advised by Profs. Gagandeep Singh, Sasa Misailovic, Minjia Zhang* Urbana, IL

- First-author of *TRAP* (NeurIPS 2025): designed an optimization-driven adversarial attack that embeds imperceptible prompts in images to reliably steer GUI-based AI agents toward attacker-chosen targets.
- Led the *Pessimistic Reward Trainer* project (ARLET @ NeurIPS 2025), addressing reward hacking in RLHF without KL regularization.
- Co-developed *SynCode* (TMLR 2025), a grammar-guided decoding framework that enforces syntactic correctness in LLM generation.
- Studied random character-based prompt augmentation attacks on LLM safety alignment (paper under review).

## INDUSTRY EXPERIENCE & COLLABORATION

**Bespoke Labs — Research Intern** *Jun 2026 — Present*

- Benchmarking frontier models' sub-agent (orchestrator) management capabilities across long-horizon tasks.
- Built a pipeline for automated data generation for long-horizon agentic tasks (200+ turns, 200K+ tokens).

**Meta Superintelligence Lab (PyTorch team)** *Aug 2025 — Feb 2026*

- Collaborated on *torchforge*, a PyTorch-native scalable RL library; co-author on two official PyTorch blog releases and contributor to the open-sourced meta-pytorch/torchforge repo. Helped scale RLVR-style training to a 512-GPU CoreWeave cluster.

**CoreWeave** *Aug 2025 — Feb 2026*

- Joint validation of distributed RL workloads on Slurm-on-Kubernetes (SUNK) with NVIDIA Quantum InfiniBand.

## SELECTED PROJECTS

**Contrastive Language Model (CLM)** *Sep 2026*

- Co-developed CLM-8B, an open-source System One model trained with contrastive learning to score candidate actions; matched Jev across computer-use, gaming, and tool-calling tasks at up to 9× lower latency.
- Project launch reached 1M views on X. blog · code · models

**Heart Disease Prediction, Hybrid CNN + BERT** *Aug 2023 — Dec 2023*

- Led 6 engineers; fused image (CNN) and tabular/text (BERT) signals to improve classification accuracy by +32% over the numeric-only baseline.

**AutoTA, Video-to-Notes Chrome Extension** *Aug 2023 — Dec 2023*

- Led 5 engineers to build a lecture-video summarization pipeline (word2vec + PEGASUS) shipped as a Chrome extension.

**Build My Portfolio, RL for Asset Allocation** *Jan 2023 — Jul 2023*

- Built a stock-trading RL environment with self-improving agents; Transformer/LSTM ticker clustering inspired by Markowitz portfolio theory.

**MindMe, Personality-Classifying Conversational Agent** *Aug 2021 — Jan 2023*

- Built a TensorFlow chatbot that infers user personality from open-ended dialogue by combining a BERT decoding head, a sentiment-analysis layer, and a CNN text-classification pipeline into a single inference stack.

## PROFESSIONAL ACTIVITIES

**Volunteer**

PLDI *2025*

**Reviewer**

ICLR *2026, 2027*
ICML *2026*
COLM *2026*
NeurIPS *2025*
AAAI *2027*

## TEACHING & LEADERSHIP

**President, Codable @ UIUC** *Spring 2024 — Spring 2025*

Led 50+ members per semester; built the org's NodeJS website used by 150+ students; ran algorithm and tech-interview workshops in partnership with CodeTree.

**Course Assistant, CS 124 (Intro to CS), UIUC** *Fall 2022*

Weekly code reviews, semester-long project mentorship, and a four-week supplementary lecture series.

**Peer Tutor, MATH 241 (Calculus III), UIUC** *Fall 2022*

Raised group average from 78.5% to 94.0% (+15.5 pts) across vector-calculus topics.

## SELECTED AWARDS

UIUC Datathon, Honorable Mention (top 10 of 200+ teams) *2025*
Illinois Engineering Outstanding Scholarship ($8000 per year) *2023, 2024*
Illinois Engineering Achievement Scholarship ($8000 per year) *2023, 2024*
HackIllinois, 2nd Place, Beginner Division. *2022*

## TECHNICAL SKILLS

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|---|---|
| **Research areas** | LLM post-training (RLHF, RLVR), agentic RL, adversarial robustness, verifier design, MLSys |
| **ML / RL stack** | PyTorch, torchforge, TorchTitan, vLLM, Monarch, TRL, VeRL, DeepSpeed, TensorFlow, JAX |
| **Infra & tooling** | CUDA, distributed training (FSDP, tensor/pipeline parallelism), Slurm, Kubernetes, AWS, Docker, Git. |